

Statistical Inference and Multivariate Analysis (MA324)

Lecture Slides

Topic 02

Simple Linear Regression

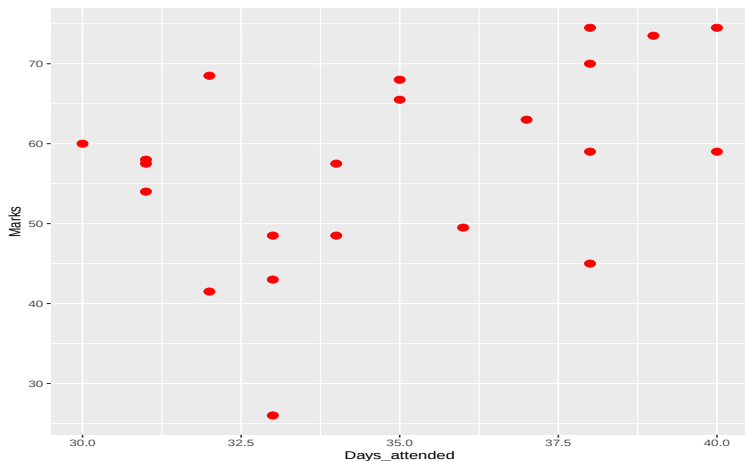
Jan-May 2026

Research Question: What is the impact of attending classes on students' final marks

Research Question: What is the impact of attending classes on students' final marks

Let's start with a real data from IITG which you can feel about it!!

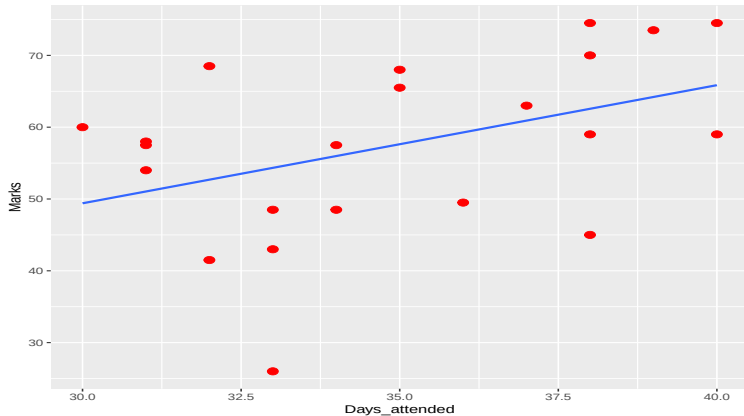
Scatter plot of number of days attended and marks by the students in MTech Data Science course (MA589)



- We will do a regression analysis to the above data to find the answer. Is using regression appropriate here?
- Strong Advice: As a Data Scientist/Statistician, you have to spent a lot of your time and effort to pre-process/clean the raw data to make it analysis ready. It is part of your job to clean the data: so learn it quickly!!
- Here is the output after fitting a regression line in R:

Fitting Regression line

Scatter plot of number of days attended and marks by the students in MTech Data Science course (MA589):
with fitted regression line



Linear Models: Simple and Multiple Linear Regressions

The regression framework can be characterized in the following way:

- We have one particular variable that we are interested in understanding or modeling, such as sales of a particular product, sale price of a home, or voting preference of a particular voter. This variable is called the target, response, or dependent variable, and is usually represented by y .
- We have a set of p other variables that we think might be useful in predicting or modeling the target variable (the price of the product, the competitor's price, and so on; or the lot size, number of bedrooms, number of bathrooms of the home, and so on; or the gender, age, income, party membership of the voter, and so on). These are called the predicting, or independent variables, and are usually represented by $x_1, x_2, \dots, x_p$.

Our Aim

- To build a relationship between y and x .
- That means to write y as a function of x .
- Observed data: (x_i, y_i) for $i = 1, 2, \dots, n$
- We want a function of x , say $f(\cdot)$, such that

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

is as small as possible.

- There are infinite solutions for this problem.
- Therefore, we want to impose constraint the set of functions over which the minimization is performed.

- Typically, a regression analysis is used for one (or more) of three purposes:
 - 1 modeling the relationship between x and y ;
 - 2 prediction of the target variable (forecasting);
 - 3 and testing of hypotheses.

Linear Models: Meaning of Linearity

What does “Linear” mean?

- A linear model is linear in the parameters β , but not necessarily in the x 's, e.g.
 - * $y = \beta_0 + \beta_1 x + \beta_2 x^2$ is a linear model, because it is linear in β (even though not in x).
 - * $y = \beta_0 + \beta_1 x^{\beta_2}$ is a non-linear model, because it is not linear in β .

Transformation

- Clearly $y = f(x) = \beta_0 x^{\beta_1}$ is not a linear model, but

$$\ln f(x) = \ln \beta_0 + \beta_1 \ln x$$

If we let $f^* = \ln f$, $\beta_0^* = \ln \beta_0$, $\beta_1^* = \beta_1$, $x^* = \ln x$, we have

$$f^* = \beta_0^* + \beta_1^* x^*,$$

which is a linear model.

Transformation

- Thus, although linear models seem to be simple and restrictive, they can actually be quite flexible by transformation of the response and the predictors.
- Linear models are not just straight lines, they can be curved. Can you give an examples?

Transform the following non-linear models to linear models:

- $y = \frac{e^{\beta x}}{1+e^{\beta x}}$, where $y \in (0, 1)$.
- $y = \frac{1}{\beta_0 + \beta_1 x_1 + \beta_2 x_2}$.

Simple Linear Regression

- Just one predictor x , i.e. $p = 1$, where p indicates the number of predictors in a multiple linear regression model.
- The model for the simple linear regression is given by

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where y is the outcome variable (random), x is the independent/predictor variable (non-random) and ϵ is the random error term. β_0 (intercept) and β_1 (slope) are model parameters (unknown constants).

- Equivalently, the model can be written for $i = 1, 2, \dots, n$ number of observations $(x_1, y_1), \dots, (x_n, y_n)$ as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, i = 1, \dots, n.$$

- How do you interpret β_0 (intercept) and β_1 (slope)?

Assumptions:

- 1 The expected value of the errors is zero: $E(\epsilon_i) = 0$ for all i
- 2 The variance of the error is constant (homoscedasticity): $Var(\epsilon_i) = \sigma^2$ for all i
- 3 The errors (ϵ_i) are uncorrelated with each other: $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$

Assumptions: What they means?

- 1 The errors (ϵ_i) are uncorrelated with each other. This violation most often occurs in data that are ordered in time (time series data), where errors that are near each other in time are often similar to each other (such time-related correlation is called autocorrelation). Violation of this assumption can lead to very misleading assessments of the strength of the regression.
- 2 The expected value of the errors is zero ($E(\epsilon_i) = 0$ for all i). That is, it cannot be true that for certain observations the model is systematically too low, while for others it is systematically too high.
- 3 The assumption of homoscedasticity (constant variance): each outcome observation (Y) has exact same variance (also for error). If Y follows Poisson distribution, this assumption is violated. Why?

Least Squares Estimation

- Goal: To estimate β_0, β_1 by minimizing error in some sense (e.g. squared error)
- One reasonable way is to use the principle of Least Squares, i.e. minimize the objective function

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

with respect to β_0, β_1 .

- Differentiate $Q(\beta_0, \beta_1)$ with respect to β_0, β_1 and equate the partial derivatives to zero to get the estimates $\hat{\beta}_0, \hat{\beta}_1$.
- The resulting equations are called normal equations:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$
$$\sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

Least Squares Estimation

- The solution is given by

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= S_{xy}/S_{xx}\end{aligned}$$

where

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \text{ and } S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are called the least squares estimator (LSE) of β_0 and β_1 respectively.

Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are linear combination of y_i 's.
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased for β_0 and β_1 , respectively.
- Variances of $\hat{\beta}_0$ and $\hat{\beta}_1$ are

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \\ \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{S_{xx}} \end{aligned}$$

Linear Estimator

Def: [Linear Estimator] An estimator $\hat{\theta}$ is called linear estimator of θ if $\hat{\theta}$ is a linear combination of random observations.

Def: [BLUE] An estimator $\hat{\theta}$ is called the best linear unbiased estimator (BLUE) of θ if $\hat{\theta}$ is linear and unbiased estimator of θ and $\hat{\theta}$ has minimum variance among all linear unbiased estimator of θ .

Gauss-Markov Theorem (Without proof)

Theorem

If the model assumptions hold true, the LSEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are BLUE of β_0 and β_1 , respectively.

A Few Definitions

- Fitted values: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, $i = 1, \dots, n$
- Residuals: Difference between observed and fitted values of y , i.e.

$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n$$

- The objective function evaluated at the LSE, i.e. $Q(\hat{\beta}_0, \hat{\beta}_1)$ is called the residual sum of squares (SS_{Res}).

$$SS_{Res} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

Properties of Least Squares Fit:

- 1 $\sum_{i=1}^n e_i = 0$. (from the first normal equation)
- 2 $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$. (using (1))
- 3 $\sum_{i=1}^n x_i e_i = 0$. (from the second normal equation)
- 4 $\sum_{i=1}^n \hat{y}_i e_i = 0$. (using (1) and (3))

Estimation of Error Variance

- It can be shown that

$$E(SS_{Res}) = (n - 2)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2}$ is called Residual variance. It's square root is called residual standard error.
- Note that SS_{Res} depends on the model. Therefore, any violation of model assumptions has serious effect on $\hat{\sigma}^2$.
- A convenient computing formula for SS_{Res} is

$$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy},$$

where

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Hypothesis Testing: β_1

- The fourth assumption of linear regression is required for testing of hypothesis: Errors are independent and normally distributed.
- Want to test the hypothesis that the slope parameter (β_1) equals to a constant (a value, say β_{10}):

$$H_0 : \beta_1 = \beta_{10} \text{ ag. } H_1 : \beta_1 \neq \beta_{10}$$

- Note that, $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ and y_i 's are independent.
- $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}}) \implies z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res}/S_{xx}}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Hypothesis Testing: β_0

- Want to test the hypothesis that the intercept parameter (β_0) equals to a constant (a value, say β_{00}):

$$H_0 : \beta_0 = \beta_{00} \text{ ag. } H_1 : \beta_0 \neq \beta_{00}$$

- $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})) \implies z = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi^2_{n-2}$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{Res}(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Interval Estimation: β_0 and β_1

- To get the CI for β_0 and β_1 , the pivots are

$$\frac{\hat{\beta}_0 - \beta_0}{\sqrt{MS_{Res}\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}}, \text{ and } \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MS_{Res}/S_{xx}}}, \text{ respectively.}$$

- A $100(1 - \alpha)\%$ CI for β_0 is

$$\left[\hat{\beta}_0 \pm t_{n-2, \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} \right].$$

- A $100(1 - \alpha)\%$ CI for β_1 is

$$\left[\hat{\beta}_1 \pm t_{n-2, \alpha/2} \sqrt{\frac{MS_{Res}}{S_{xx}}} \right].$$

Interval Estimation: CI for σ^2

- To get the CI for σ^2 , the pivots is

$$\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$$

- A $100(1 - \alpha)\%$ CI for σ^2 is

$$\left[\frac{(n-2)MS_{Res}}{\chi_{n-2;\alpha/2}^2}, \frac{(n-2)MS_{Res}}{\chi_{n-2;1-\alpha/2}^2} \right].$$

Interval Estimation: CI for mean response

- A regression model can be used to estimate the mean response $E(y)$ for a particular value of the regressor variable x . Let x_0 be a value of the regressor variable (must be with the range of original data on x). Then $E(y|x_0) = \beta_0 + \beta_1 x_0$.
- Then, $\hat{y}_0 = \widehat{E(y|x_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0$
- And, $\hat{y}_0 \sim N\left(\beta_0 + \beta_1 x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)$
- Pivot: $\frac{\hat{y}_0 - (\beta_0 + \beta_1 x_0)}{\sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}} \sim t_{n-2}$
- A $100(1 - \alpha)\%$ CI for $\beta_0 + \beta_1 x_0$ is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Prediction Interval for New Observation:

- Let x_0 be a value of the regressor variable (must be with the range of original data on x).
- The true value of the response is y_0 (corresponding to x_0).
- We want to provide an interval I such that $P(y_0 \in I) = 1 - \alpha$
- Note that the point estimate of y_0 is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
- Consider $\psi = y_0 - \hat{y}_0$.
- Then, $E(\psi) = 0, Var(\psi) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$
- $\psi \sim N \left(0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \right)$
- Pivot: $\frac{y_0 - \hat{y}_0}{\sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}} \sim t_{n-2}$

Prediction Interval for New Observation:

- A $100(1 - \alpha)\%$ prediction interval is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Coefficient of determination: R^2

- Coefficient of determination:

$$R^2 = \frac{SS_{Reg}}{SS_T} = 1 - \frac{SS_{Res}}{SS_T} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where $SS_{Reg} = \sum_i (\hat{y}_i - \bar{y})^2$

- It is a bounded quantity: $0 \leq R^2 \leq 1$
- R^2 is interpreted as the “percentage of variation explained by the model”.
- Higher values of R^2 are desirable (R^2 close to 1 indicates a good fit), but “how high is high” depends on the context.
- In multiple linear regression, adding a variable to a model can increase the value of R^2 . To overcome this problem:

$$\text{Adjusted } R^2 : \quad R_{adj}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2 / (n - p - 1)}{\sum_i (y_i - \bar{y})^2 / (n - 1)}.$$

F -Test for Regression

- Large value of the sum of squares for regression $SS_{Reg} = SS_T - SS_{Res}$ indicates the simple regression mean function $E(Y|X = x) = \beta_0 + \beta_1 x$ should be a significant improvement over the mean function $E(y|X = x) = \beta_0$.
- This is equivalent to saying that the additional parameter (β_1) in the simple regression mean function is non-zero. In other words, that $E(Y|X = x)$ is not constant with respect to X .
- To test

$$H_0 : E(Y|X = x) = \beta_0 \text{ ag. } H_1 : E(Y|X = x) = \beta_0 + \beta_1 x$$

- Test statistics is a ratio, defined as F :

$$F = \frac{SS_{Reg}/1}{\hat{\sigma}^2} = \frac{SS_{Reg}/1}{SS_{Res}/(n-2)} \sim F_{1,n-2}$$

Simple Linear Regression in Heights Data

Heights Data Example

- Data on heights of $n = 1375$ mothers in the UK under the age of 65 and one of their adult daughters over the age of 18.
- Collected and organized during the period 1893–1898 by the famous statistician Karl Pearson.
- The dataset is publicly available
 - R-packages: alr4, UsingR
 - GitHub: https://github.com/kbroman/PearsonData/blob/master/mother_daughter.csv
- A historical use of regression to study inheritance of height from generation to generation
- Sample correlation coefficient is 0.491

Simple Linear Regression: Using R

```
> height.lm <- lm(Dheight~Mheight,data=heights)
> summary(height.lm)
```

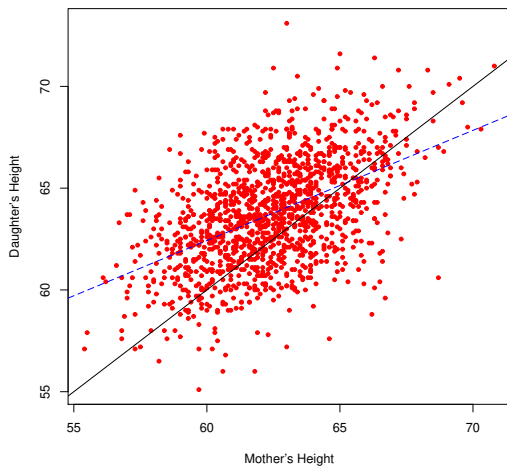
...

Coefficients:

	Estimate	Std.Error	t-value	Pr(> t)
(Intercept)	29.91744	1.62247	18.44	<2e-16 ***
Mheight	0.54175	0.02596	20.87	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.266 on 1373 degrees of freedom
Multiple R-squared: 0.2408,
adjusted R-squared: 0.2402
F-statistic: 435.5 on 1 and 1373 DF,
p-value: < 2.2e-16



Regression Effect

- It's an empirical phenomenon, also called “regression to the mean” or “regression to mediocrity”, e.g.
 - Daughters of tall mothers **tend to be tall, but not as tall** as their mothers; daughters of short mothers tend to be short, but not as short as their mothers (same trend between sons and fathers)
 - Students **doing well in Mid-term tend to do well in the Final, but not as well** as the Mid-term; students doing poorly in Mid-term tend to do poorly in the Final, but not as poorly as the Mid-term!

Caution

- Do you think a moderate or high absolute value of correlation coefficient makes a data ideal for LR?

Caution

- Do you think a moderate or high absolute value of correlation coefficient makes a data ideal for LR?
- Consider datasets given in next slide.

Anscombe's Examples (1973)

Dataset 1		Dataset 2		Dataset 3		Dataset 4	
x	y	x	y	x	y	x	y
10.00	8.04	10.00	9.14	10.00	7.46	8.00	6.58
8.00	6.95	8.00	8.14	8.00	6.77	8.00	5.76
13.00	7.58	13.00	8.74	13.00	12.74	8.00	7.71
9.00	8.81	9.00	8.77	9.00	7.11	8.00	8.84
11.00	8.33	11.00	9.26	11.00	7.81	8.00	8.47
14.00	9.96	14.00	8.10	14.00	8.84	8.00	7.04
6.00	7.24	6.00	6.13	6.00	6.08	8.00	5.25
4.00	4.26	4.00	3.10	4.00	5.39	19.00	12.50
12.00	10.84	12.00	9.13	12.00	8.15	8.00	5.56
7.00	4.82	7.00	7.26	7.00	6.42	8.00	7.91
5.00	5.68	5.00	4.74	5.00	5.73	8.00	6.89

Means & Standard Deviations

Data Set	N	<i>x</i>		<i>y</i>	
		Mean	SD	Mean	SD
1	11	7.50	2.03	9.00	3.32
2	11	7.50	2.03	9.00	3.32
3	11	7.50	2.03	9.00	3.32
4	11	7.50	2.03	9.00	3.32

The means and SDs for the 4 data sets are identical to two decimals.

Correlations between x and y

Data Set	Pearson r	R-squared	Adj. R-sq
1	0.82	0.67	0.63
2	0.82	0.67	0.63
3	0.82	0.67	0.63
4	0.82	0.67	0.63

Correlations, R-squared, adjusted R-squared, are all identical to two decimals.

ANOVA Summary Tables

Data Set	Source	SS	df	MS	F	p -value
1	Regression	27.490	1	27.490	18.003	0.002
	Residual	13.742	9	1.527		
	Total	41.232	10			
2	Regression	27.470	1	27.470	17.972	0.002
	Residual	13.756	9	1.528		
	Total	41.226	10			
3	Regression	27.500	1	27.500	17.966	0.002
	Residual	13.776	9	1.531		
	Total	41.276	10			
4	Regression	27.510	1	27.510	17.990	0.002
	Residual	13.763	9	1.529		
	Total	41.273	10			

The Regression Coefficients

Data Set		Estimate	SE	t	p -value	95% CI	
						Lower	Upper
1	Constant	3.00	1.124	2.67	0.026	0.459	5.544
	x	0.50	0.118	4.24	0.002	0.233	0.766
2	Constant	3.00	1.124	2.67	0.026	0.459	5.546
	x	0.50	0.118	4.24	0.002	0.233	0.766
3	Constant	3.00	1.125	2.67	0.026	0.455	5.547
	x	0.50	0.118	4.24	0.002	0.233	0.767
4	Constant	3.00	1.125	2.67	0.026	0.456	5.544
	x	0.50	0.118	4.24	0.002	0.233	0.767

For all 4 models, $\hat{y} = 0.5x + 3$

Which dataset is best described by the model?

- Judging by everything we've just seen, it appears that the models are all equally good
- But if that were true, I wouldn't be doing this talk!
- Let's look at scatter plots between x and y

Importance of graphing data before analyzing it

